A BLOWUP PULLS BACK EFFECTIVE CARTIER DIVISORS AND MEROMORPHIC FUNCTIONS

THE STACKS PROJECT AUTHORS

Editorial note. All delivered mathematical text is the author's own, copied from the pinned Stacks Project chapter file divisors.tex; no statement or proof step was rewritten, reordered or removed.

Editorial note. Carrier window: statement and proof both lie inside the window 8000-10000 granted to this lane by CONTROL 8.83.

Editorial note. The proof's algebra fact is proved in place, so the card is self-contained apart from the two reduction lemmas recorded as navigation pointers.

Editorial note. Preamble: sources/stacks-divisors/retained-preamble.tex (88 lines, verbatim present).

Allowed standard prerequisites (editorial).

- Local description of a blowup by affine blowup algebras. Lemma lemma-blowing-up-affine, used at proof lines 8577-8579. Reduces the assertion to the affine case.
- Local characterisation of effective Cartier divisors. Lemma lemma-characterize-effective-Cartier-divisor, used at proof lines 8577-8579. Turns the pullback statement into the nonzerodivisor statement.
- Definitions of the pullback of an effective Cartier divisor and of meromorphic sections. Definitions definition-pullback-effective-Cartier-divisor and definition-pullback-meromorphic-sections, used at proof lines 8570-8574. Fix what has to be shown defined.

Source reference. Original definition 01WV: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 2496–2496. Navigation pointer into the Stacks Project; the standard fact it supplies is applied inside the counted proof span and the argument resting on it stays there.

Source reference. Original definition 02OT: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 3528–3528. Navigation pointer into the Stacks Project; the standard fact it supplies is applied inside the counted proof span and the argument resting on it stays there.

Source reference. Original lemma 0804: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 10169–10169. Navigation pointer into the Stacks Project; the standard fact it supplies is applied inside the counted proof span and the argument resting on it stays there.

Source reference. Original lemma 01WS: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 2493–2493. Navigation pointer into the Stacks Project; the standard fact it supplies is applied inside the counted proof span and the argument resting on it stays there.

Date: stacks-project:divisors.tex:local-snapshot-20261012; GNU FDL 1.2 or later.

Lemma 0.2. *Let X be a scheme. Let $b : X' \to X$ be a blowup of X in a closed subscheme. The pullback $b^{-1}D$ is defined for all effective Cartier divisors $D \subset X$ and pullbacks of meromorphic functions are defined for b (Definitions 01WV and 02OT).*

Proof. By Lemmas 0804 and 01WS this reduces to the following algebra fact: Let A be a ring, $I \subset A$ an ideal, $a \in I$, and $x \in A$ a nonzerodivisor. Then the image of x in $A[\frac{I}{a}]$ is a nonzerodivisor. Namely, suppose that $x(y/a^n) = 0$ in $A[\frac{I}{a}]$. Then $a^m xy = 0$ in A for some m. Hence $a^m y = 0$ as x is a nonzerodivisor. Whence y/a^n is zero in $A[\frac{I}{a}]$ as desired. □